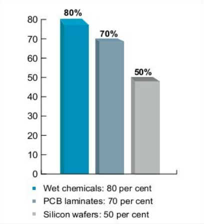

Fig. 3: The top three demand generating categories of electronic chemicals

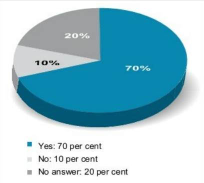

Fig. 4: Does a demand-supply gap exist in the market?

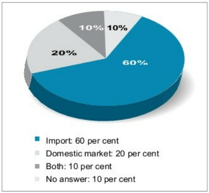

Fig. 5: Major sourcing channels for electronic chemicals and materials

expected an even more accelerated growth of over 30 per cent.

Market segments

The materials used in electronics manufacturing are well-known high-purity chemicals such as photoresists, wet chemicals, acids, gases and solvents, as well as wafers and laminates.

Market challenges

- Miniaturisation of electronic components and products has resulted in lesser use of chemicals
- Cost sensitivity of the markets
- Limited availability of high-quality products
- Limited scale of electronics manufacturing in the country
- High import costs
- Unfavourable tax and duty structures
- Lack of quality logistics support or services, resulting in poor handling and wastage
- Inefficient supply chain

CHEMICALS AND MATERIALS USED IN ELECTRONICS MANUFACTURING

Type	Application	Form
• Silicon wafers • PCB laminates • Specialty gases • Photoresists • Wet chemicals • CMP slurries • Low-K dielectric • Others	• Semiconductor packaging • Integrated circuits (IC) manufacturing • PCB manufacturing	• Solid • Liquid • Gaseous

Based on the types of products, the market can be divided into eight segments: silicon wafers, PCB laminates, photoresists, specialty gases, CMP slurries, wet chemicals, Low-K dielectrics and others (refer the table).

Market demand

As stated earlier, silicon wafers constitute the major chunk of the global electronic chemicals and materials market, and are widely used as base materials in semiconductors' manufacturing. PCB laminates are widely used for PCB manufacturing. Low-K dielectrics are relatively recent applications of semiconductors and ICs, and therefore are expected to grow aggressively in the next five years across the globe.

The Indian market for electronic chemicals and materials does not follow global trends. It is expected to be primarily driven by the demand for wet chemicals and PCB laminates. Environmentally-safe products will also be in demand. Survey participants predict a huge demand for wet chemicals. According to them, future growth will predominantly come from the following top three demand generating categories (Fig. 3):

1. Wet chemicals
2. PCB laminates
3. Silicone chemicals

Moreover, there is a demand for plain PET and PP films, liquid and powder epoxy, TPCS and TCA wire.

Demand-supply gap

In India, the demand for electronic chemicals and materials is primarily met through higher imports in the absence of significant local production. Seventy per cent of survey participants cited demand-supply gap on account of inadequate local production (Fig. 4). Value addition by local manufacturers is quite limited, as most high-value and critical materials are still imported.

Sixty per cent survey participants mentioned that imports are the main source of electronic chemicals and materials used in India. However, 20 per cent participants felt local production is adequate to meet the demand, while 10 per cent said that demand is met by both local production and imports (Fig. 5).

The local production ecosystem for electronic chemicals and materials is still not mature enough. Therefore, currently, local sourcing is primarily limited to:

1. Adhesives
2. Basic-level PCB cleaning and coating solutions